The Most Important Rule

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Good morning, ladies and gentlemen.

I thank the Sphinx Club for inviting me to speak. I am very honored that they would extend me such an invitation.

I also thank everyone here today—students, faculty, staff, guests—for coming to hear my humble remarks. Some of you may have big expectations. I will try not to disappoint you.

Last, but certainly not least, I thank the many, many people who—both knowingly and not—have helped me develop this talk. In particular, I thank Professors J.D. Phillips, Joe O'Rourke, and Mark Brouwer; and Dr. Keith Baird.

I am sure my topic comes as no surprise to those who know me well. I like rules. I like certainty. I like discipline. I like order, and I abhor chaos.

This desire for order probably drew me to be a scientist, and in particular to study physics, mathematics, and perhaps even computer science. I want to believe that this great chess game we call the universe has rules by which it operates. That's what scientists do—observe the game being played by others to deduce what rules govern the game. They may even try to play themselves to see what moves are allowed. Scientists all believe—whether they admit it or not—that this empirical evidence is the best way to understand how the world actually is. They try to use empirical evidence to explain the universal and necessary connections of cause and effect. Hume argues that we have no evidence that cause and effect actually exist, only constant conjunction. To Hume, I say I choose to believe in cause and effect because to assume otherwise is to concede that there is no order, which I find very distasteful.

I studied physics in college because I want to know those rules. I want to know why objects behave the way they do. I want to understand matter, motion, space, and time. If you release an object, why does it fall to the ground? If you are playing billiards, how must you hit the cue-ball to get the eight-ball into the corner pocket? How do sound and light travel? How does a steam engine work? I want to know the underlying principles of all these and so much more.

Mathematics attracted me first by its applications to physics and the ability of applied mathematics to actually calculate formulae that describe physical laws and predict future events. Later, the allure of abstract mathematics and its categorizing mathematical structures so we can prove theorems—or rules—about their behavior caught my eye. What are the essential properties of the real numbers and polynomials that cause our manipulations of the two structures to be so similar? What can we prove about any structures that have these same properties? Which of these properties are necessary for these results? If I remove a property, what remains? If I add a different property, what new theorems result? I fought the entrancing spell of abstract mathematics for years, believing that it held no applications, especially to physics and the world of natural order. I did not see its order as real. Eventually, I came see pure mathematics held far more order than physics, and it was at least as real. When I finally read G. H. Hardy's *A Mathematician's Apology* after graduate school, I immediately connected with his description of a mathematician being in much more direct contact with reality than a physicist. As Hardy says (§24):

> [T]his realistic view is much more plausible of mathematical than of physical reality, because mathematical objects are so much more than what they seem. A chair or a star is not in the least like what it seems to be; the more we think of it, the fuzzier its outlines become in the haze of sensation which surrounds it; but '2' or '317' has nothing to do with sensation, and its properties stand out the more clearly the more closely we scrutinize it.

In mathematics, I could know the rules of the game, whereas physics only offered guesses and approximations. One never knows for sure what rules—if any—truly describe the universe.

I can even see my particular area of research—symbolic computation—fitting into this general theme. It combines abstract mathematics with the calculating power of computers. Unlike its cousin field, numerical analysis, symbolic computation uses mathematical structures to allow exact computations. No messy floating point numbers. No ugly round-off errors. No worrying about the stability of an algorithm or the condition number of a matrix. No trying to get close to the answer. You get the answer.

I find this same desire for rules elsewhere in my life. For example, I like formalities. Living in the south for a while, I found the way children are raised to address their elders by "Mister", "Miss", and "Missus"—as well as to say "Sir" and "Ma'am"—very appealing. I do not hear it much in Indiana, or anywhere else I have lived outside of the south. I think our culture, where we call everybody by their first names, has lost something very important. In our attempt to be more intimate with everyone, we have lost true intimacy. In our quest to be friends to everyone, we have lost true friendship. Once, we reserved the use of first names for only close friends, but it no longer means anything for me to allow somebody to call me by my first name. This is no longer something reserved for close friends. I no longer even have the ability to grant this privilege. The privilege no longer exists. Now everybody acts as if he or she is my best buddy—from professionals I meet around town to the solicitor who calls me on the telephone. This only cheapens that relationship so that nobody is that close anymore. Confucius was right about filial piety, and I am happy to see Wabash College has not completely conceded defeat to this movement yet.

Yes, I like rules, but it is more than just a personal preference. Rules are necessary. Without rules, society could not function. We have a rule that we drive on the right side of the road. Why? So we don't kill each other whenever we go out on the highway. We have rules protecting private property rights and governing communal property to avoid the "tragedy of the commons''. We have rules against murder, theft, libel, slander, using drugs, drinking alcohol, smoking tobacco products. We have rules protecting us against certain acts, and against us protecting ourselves against other acts. We have rules about licenses to obtain and taxes to pay; about identification to be shown to buy certain products, to cross national borders, to board an airplane, and to exercise your constitutional right to vote. Some places now have rules against swearing in bars, eating foie gras, and maybe soon against gaining too much weight.

Here in Indiana, one of the biggest controversies today comes from the rules governing standardized time—or as my brother likes to archaically call it: "railroad time". (You would understand if you knew my little brother.) Standardized time was introduced in the nineteenth century with the growth of the railroads. It eliminated scheduling confusion that resulted from each stop using its local—or solar—time. Each town used to set its own time based upon the sun being at its highest point at noon. Before railroads and their faster travel—as well as before telephones and instantaneous long-distance communication—this difference in time was not so noticeable to most people that it bothered them, especially before accurate timepieces became more common place. However, with the railroads came the need to publish accurate timetables showing exactly when a train should arrive and depart every station. At first, each railroad kept its own standardized time, but this meant a passenger making a trip involving several connections had to keep track of the time for each railroad. Even more tragically, it meant trains from different companies could be on the same track, both conductors thinking the track was clear, because the two used different times. Such a situation arose in New England in 1853 with fatal consequences, which lead to a regional standard time. Standardized time zones eventually took shape around the world throughout the nineteenth century. In October 1883, the heads of the major railroads met in Chicago and adopted the Standard Time System, which is essentially the time zones we currently use. Most states adopted it by the end of the year, despite much local opposition. An Indianapolis newspaper report of 17 November 1883 protested that people would have to "eat, sleep, work, … and marry by railroad time". The United States government formally adopted it in 1918, in an act that also introduced Daylight Savings Time.

Perhaps the least-understood category of rules, and one with which I have had a long fascination, is parliamentary law. In the United States, the most famous set of these came about from the experiences of a U.S. army engineering officer during and after the civil war (or the war of Northern Aggression, as some still call it in the south). In 1863, following a recurrence of tropical fever, this twenty-five year-old officer was transferred to New Bedford, Massachusetts. While there, he was asked to preside over a meeting, a request that he felt he was duty-bound to accept. Afterward, he decided he must learn some parliamentary procedure. He found his first "rules for deliberative assemblies" on a few pages in a book on another subject, and he copied these rules onto a piece of paper that he carried with him for several years. After the war, he was transferred to San Francisco in 1867. There, he worked with people from many different parts of the country in several organizations seeking to improve social conditions in this turbulent community. Each member of these organizations brought different, and often very strong, convictions of parliamentary rules. This officer decided to research the subject further. After obtaining several parliamentary manuals, including one written by Thomas Jefferson for the U.S. House of Representatives, he decided to write his own short manual, of perhaps sixteen pages, that the organizations to which he belonged could use. However, he was transferred first to Portland, Oregon, in 1871, and military engineering duties kept him too busy to work on his project until January 1874, in Milwaukee, Wisconsin, when severe winter weather closed the harbor and gave him three months free. After that, his duties forced his work to proceed slowly, and it was not until the end of 1875 that Major Henry Martyn Robert saw the first complete printing—at his personal expense—of his *Pocket Manual of Rules of Order for Deliberative Assemblies*, which by then had grown to 176 pages!

This first edition of *Robert's Rules of Order* was published in February 1876. Before he died in 1923, General Robert personally oversaw the publication of the first four editions. The last of these was a complete revision and large expansion of his original rules, titled *Robert's Rules of Order Revised*, in 1915. A second complete rewriting followed in 1970, doubling the size of the sixth edition and recasting the manual to be self-explanatory. The current tenth edition of *Robert's Rules of Order Newly Revised* has over 700 pages. By now *Robert's Rules* has become a staple in the United States and Canada. Although it is certainly not the only parliamentary authority, it is perhaps the most referenced in non-legislative bodies in this country.

As with all systems of parliamentary procedure, *Robert's Rules* details a set of rules for a deliberative assembly relating to the orderly transaction of business in meetings and to the duties of the assembly's officers. Parliamentary law aims to facilitate the smooth functioning of the assembly and to provide a firm basis for resolving questions of procedure that arise.

In the English-speaking world, parliamentary law derives from rules that developed over the centuries in the English parliament through a continuing process of decisions and precedents, much like the growth of common law. This evolutionary development represents a movement away from making decisions based on "consensus", in its original sense of unanimous agreement, and toward a decision by majority-vote system we know today. This evolution was spurred by a recognition that requiring unanimity or near unanimity can become a form a tyranny. Trying to make such a requirement the norm can lead to decisions that appear to be unanimous but with which many members disagree, perhaps strongly. Pressure from a leader, an apparent majority, or even a very vocal minority, can silence other members' voices and easily lead to the assembly making decisions based only on a pseudo-consensus. Members can grow angry and resentful, believing—wrongly—that everybody else agrees with the decision made by the assembly. Animosity can grow, and groups can fracture. Modern parliamentary law is based on the belief that majority voting along with lucid and clarifying debate—even verbal combat—resulting in a decision representing the view of the majority—far more clearly discerns and demonstrates the assembly's will.

The sixteenth and seventeenth centuries, in particular, were a turbulent time in England. The kingdom went from Catholicism to Protestantism, back to Catholicism, and finally back to Protestantism. It was ruled by a king, a protector, and two queens. The king of Scotland became the king of England, uniting the crowns. His son ruled both kingdoms until he lost his head. An English republic was declared. A Lord Protector took over. He conquered Scotland and created a united commonwealth. The monarchy was restored—and with it, two separate kingdoms. The king was overthrown in a "bloodless" revolution that was not really bloodless. The bill of rights was passed. English parliamentary democracy was born, and eventually the two kingdoms where united into one.

This time also saw many advances that we would recognize today as integral components of parliamentary law. Perhaps not surprisingly, parliament had a prolonged internal conflict over its prerogatives—as opposed to those of the king—which stimulated increased interest in procedural matters, especially in the House of Commons. These advances included the rule to discuss only one subject at a time in 1581, the rule to alternate between opposing points of view when assigning the floor in 1592, the requirement that the chair always call for the negative vote—and not just the positive—in 1604, the requirement of decorum and avoiding personalities in debate also in 1604, confining debate to the merits of the pending question in 1610, and the ability to divide a question in 1640.

One of the most important advances, perhaps, was the formation of the *Journal of the House of Commons* around 1550, which became an established source of precedent in matters of procedure. At roughly the same time, Sir Thomas Smyth prepared the first formal treatment of the Commons' procedure, and in 1689, George Petyt published a small pocket manual for members of parliament to reference. Before then, establishing precedent usually meant sifting through the *Journal of the House of Commons*, which I imagine to be much the same procedure the college faculty uses to find precedents to this day. (If only we had our own George Petyt.)

The seventeenth century also saw the first permanent English colonies in the new world. The colonists brought these rules of order with them, and modified them to meet their particular needs. Eventually, after two-hundred years of development in the western hemisphere, Major Robert sought to simplify and codify these rules in a simple manual that anybody could use. Common sense was to rule the day.

Many people complain about parliamentary procedure and *Robert's Rules*. They say it is too difficult to understand, too unwieldy to use. I think understanding the underlying principles upon which *Robert's Rules* is based can greatly alleviate these complaints. The first of these principles is to protect the rights of individuals and subgroups within the assembly's total membership:

- of the majority,
- of the minority, especially a strong minority,
- of the individual members,
- of absentees, and
- of all these together.

The protection of these rights appears throughout *Robert's Rules*. That all debate must be about a motion on the floor protects everyone's rights by keeping the discussion focused.
Every member of the assembly has the right to bring a motion to the floor, although some motions require previous notice to allow absent members to have their voices heard. Requiring a second protects the majority from debating measures in which only a single member is interested. A second signifies another member is also interested in the topic; it does not necessarily mean anybody else is in favor of it. Perhaps somebody wants to debate it to vote it down once and for all. Ultimately, the majority decides the general will of the assembly, but only following a full and free discussion. The right of a minority or an individual member to this discussion can only be denied by at least two-thirds of those present and voting. A two-thirds vote is also required to modify the rules of order. Once the motion is moved, seconded, and read by the chair, it belongs to the assembly, and not to any particular member. Any amendments must be approved by the majority of the assembly. *Robert's Rules* does not recognize a "friendly

amendment". This protects the majority from an individual or a minority changing the pending question.

Another principle used by *Robert's Rules* is the protection from instability, which can arise from such factors as a small variation in attendance. For this reason *Robert's Rules* requires previous notice for some actions and places a greater requirement on changing a previous action than on taking the action in the first place. Motions can be rescinded, repealed, or annulled, but only if moved by a member who voted on the prevailing side, and it takes at least two-thirds of those voting and present, if not a majority of the entire membership, unless previous notice is given.

Even if one understands the principles on which *Robert's Rules* is based, complaints may still remain: it is too pedantic, too formal. It makes our meetings run less smoothly. Because the entire idea behind parliamentary law is to facilitate the smooth functioning of the assembly, this is a big problem. Luckily for us, there is a rule to govern it: the very first rule! Section 1 of *Robert's Rules of Order Newly Revised*, pages one and two, defines the type of gathering to which parliamentary law applies: the deliberative assembly. In its definition, it lists seven distinguishing characteristics of the deliberative assembly. Any gathering that does not meet all seven is not a deliberative assembly, and thus *Robert's Rules* is not appropriate for it. Most of the characteristics deal with the group having the freedom to determine—in full and free discussion—courses of action in the name of the entire membership, even those absent, the members of the group having the freedom to vote as they chose, each member's vote having equal weight, et cetera. However, to me, the third characteristic is key: "The group must be of such size that a degree of formality is necessary in its proceedings". In other words, if a group is small enough, it may not need the degree of formality that parliamentary procedure provides. This formality, in fact, may hamper its deliberations. *Robert's Rules* is not designed for this group. If a rule is too formal, if its strict observance hampers rather than facilitates the smooth functioning of the assembly, it should not be enforced.

This is the most important rule. It helps define, and thus bound, *Robert's Rules*. *Robert's Rules*—like all rules, even like science—has boundaries. To apply a set of rules, you must know to what they apply. You must know their definitions—their boundaries. Just as the absence of rules can lead to tragedy, so can obsessive devotion to rules go too far. Should an eighth-grader be stripped of his class office and suspended from school for having Skittles on school grounds? Should a six-year old be suspended for a haircut that is too short? Should a fifteen-year-old be given detention for stopping a run-away bus full of students and possibly saving their lives? Should thousands of political refugees be turned away because of strict immigration quotas, sent back to almost certain death in a country that publicly announced it had a "problem" with these people, a problem that needed a "final solution"? Should "I was just following orders"—"I was just following the rules"—be a valid excuse?

The U.S. military responds by requiring servicemen and –women to obey all "lawful orders", but what makes an order lawful? On what authority does a rule stand? One way to answer this question is to base it on another rule. On what authority does this new rule stand? Define yet another rule! You can continue making rules about rules ad infinitum. You can have turtles all the way down. Eventually you just have to start with something—be it axioms, deductive apparatus, faith, et cetera. This first rule of *Robert's Rules* is necessarily vague—just as with another rule we all know. Its vagueness is its strength.

Mathematicians tend to take the other route: to try to define very specific rules on which to base everything else. In 1900, at the second International Congress of Mathematicians in Paris, David Hilbert famously posed 23 questions that set the course for much of the mathematical research of the coming century. His second question essentially asked for a consistent set of axioms—rules—to describe all of arithmetic. Thirty-one years later, Kurt Gödel answered the question in the negative. He argues that any system of axioms—any system of rules—about arithmetic cannot be both complete and consistent. In other words, either it must leave some statement unproved, or it must prove some statement both true and false. Some statement is undecidable.

Kant recognizes that Hume's science is not self-consistent. One cannot use empirical evidence to answer the question of whether empirical evidence is the best way to know the world because to do so already assumes the answer to the question. Science cannot answer the question of whether science is the best way to proceed—whether cause and effect exist. In fact, Kant argues that causality is not derived from experience, but experience is derived from causality. One can either do as Kant does and try to ground science in some other discipline—such as metaphysics—or one can do as I think most scientists do—as I do—and take it as a matter of faith.

Thank you.